MIL-T-52877B
20 February 1980
SUPERSEDING
MIL-T-52877A(ME)
9 May 1977

MILITARY SPECIFICATION

TRANSMISSION GROUP DATA:

TRANSMITTER, DATA: T-1257/FSS-9(V)

AND

RECEIVER, DATA: R-1861/FSS-9(V)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a data transmission group for the Joint-Services Interior Intrusion Detection System (J-SIIDS). The data transmission group consists of data transmitters, receivers, and matching code plug sets for reporting control unit status to a monitor cabinet.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.

FSC 6350

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-566	- Boxes, Folding, Paper Board.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-C-45662	- Calibration System Requirements.
MIL-C-52853	- Control Unit, Alarm Set: C-9412/FSS-9(V).
MIL-C-52876	- Cabinets, Monitor, CY-7359/FSS-9(V), CY-7360/FSS-9(V) and CY-7361/FSS-9(V).
MIL-A-52878	- Alarm-Monitor Group.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781	- Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13220E3500	- Receiver, Data: R-1861/FSS-9(V).
TA13220E3600	- Transmitter, Data: T-1257/FSS-9(V).

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D13220E3607 - Code Plug.

(Copies of specifications, standards, and drawings required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The data transmission group shall be in accordance with Top Assemblies TA13220E3500 and TA13220E3600, and code plugs shall be in accordance with Drawing D13220E3607, and as specified herein.

3.1.1 Major components. The major components of the data transmission group are as follows:

(a) Data transmitter - TA13220E3600.
(b) Data receiver - TA13220E3500.
(c) Code plug sets - D13220E3607.

3.1.2 Function. The data transmission group is an integral part of the Joint-Services Interior Intrusion Detection System (J-SIIDS). The group provides a secure method of communicating the status at the control unit conforming to MIL-C-52853 to the monitor cabinet conforming to MIL-C-52876. The transmitter encodes the status at the control unit then transmits this status to the receiver at the monitor cabinet where the status is decoded and causes appropriate status or alarm conditions to be displayed on a monitor module conforming to MIL-A-52878.

3.1.3 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the data transmitter and receiver. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

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3.2 First article (preproduction model). The contractor shall furnish one or more data transmission groups for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce data transmission groups that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more data transmission groups for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.8).

3.5 Environmental conditions. Each data transmission group shall withstand individual environmental conditions specified herein, or combinations thereof, without damage (see 6.4.2), or malfunction when tested as specified in 4.6.2.5.

3.5.1 Operational.

(a) Temperature range of -20° F to +150° F.
(b) Humidity up to 95 percent relative.

3.5.2 Nonoperational and storage.

(a) Temperature range: -30° F to +165° F.
(b) Humidity: up to 95 percent relative.
(c) Shock: 20 times the acceleration due to gravity (g's) for 11 milliseconds (ms) duration and bench handling.
(d) Vibration: transportation conditions.

3.6 Detailed technical requirements.

3.6.1 Transmitter. When the input to the transmitter is connected as

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specified in Table I, the transmitter shall process and utilize the input functions listed. After processing these input signals, the transmitter shall transmit the resulting output signal to the receiver by means of a 600 ohm balanced transmission line.

3.6.1.1 Input connections. Table I lists the input connections to the transmitter.

Table I. Input connections

Lead color	Identification of lead	Function
Blue	1-I	Instantaneous ALARM/NO ALARM input.
Orange	1-D	Latched ALARM/NO ALARM input.
Yellow	2	ACCESS-SECURE input.
White	3	AC POWER ON/AC POWER FAIL input.
Gray	4	Common return for status inputs.
Red	5	+20 volts direct current (Vdc) input.
Black	6	20 Vdc return (ground).
Green	7	Resynchronization (resync) initiation.

3.6.1.2 Output connections. The output from the transmitter shall be by means of a two-terminal (captive 5-40 screw) barrier strip.

3.6.1.3 Code plugs. Provision shall be made in the transmitter to house a code plug conforming to Drawing D13220E3607, the selection and use of which shall permit altering the pseudorandom output bit pattern generated by the transmitter. By selection and use of a suitably wired code plug, one of 16 unique maximum length cipher bit patterns shall be generated by the transmitter. Provision shall be made in the receiver to house a companion (identically wired) code plug which will enable proper decoding of the encoded alarm and status signals from the transmitter. Neither the code plug in the transmitter nor that in the receiver shall be accessible without removing the covers to the units. Unless otherwise specified (see 6.2), code plugs shall not be furnished as part of the data transmission group. Code plugs when furnished, shall be furnished in matched sets (identically wired) of two or more plugs, as specified (see 6.2).

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3.6.1.4 Output data word. The output data words from the transmitter to the receiver shall consist of a serial stream of 12-bit words occurring at a word rate of one per second (word rate = 1). These output words shall be generated in two configurations, one being a normal data word and the other being a unique word used to synchronize the receiver to the transmitter. Figure 1 illustrates the normal data word, and Figure 2 illustrates the unique word that shall be generated in the transmitter. The unique word shall be generated upon initial application of operating power to the transmitter or whenever the resync initiation lead (green) is momentarily grounded. The unique word shall be transmitted as the fifth word in the data stream following the occurrence of either event above. The first four words prior to the unique word shall appear as normal data words with Bit 6 (Access/Secure bit) set to Access. After the occurrence of the unique word, transmission of the normal words shall resume with all status and alarm bits set to their true condition. Enciphering shall be provided within the transmitter logic to cause the format of Bits 5, 6, 8, 9, and 11 to randomly change every five data words (every 5 seconds). Bits 9 and 11 (cipher bits) shall always be opposite (i.e., if Bit 9 is at a "1" state, Bit 11 will be at a "0" state and vice versa). Bit 12 shall always be at a state to make the parity of each word even.

3.6.1.5 Function.

3.6.1.5.1 Input. The input functional tolerances shall be as listed in Table II.

Table II. Input status functional tolerances

Condition	Impedance level at input
ALARM I (alarm instantaneous)	Greater than 100 kilohms (k ohms).
NO ALARM I (no alarm instantaneous)	Less than 2k ohms.
ALARM L (alarm latched)	Greater than 100k ohms.
NO ALARM L (no alarm latched)	Less than 2k ohms.
ACCESS	Less than 2k ohms.
SECURE	Greater than 100k ohms.
AC POWER FAIL	Greater than 100k ohms.
AC POWER ON	Less than 2k ohms.

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3.6.1.5.2 Output. The output functional requirements shall be as listed in Table III.

Table III. Output functional requirements

Output function	Definition
Number of outputs	1
Output level	Selectable, 0 to -12 dBm in nominal 3 dBm steps.
Output impedance	600 ohms balanced.
Bit rate	12 Hertz (Hz).
Type of transmission	Monologue, time division multiplex (TDM), frequency shift keyed (FSK), 12-bit code.
FSK logic tones:	1070 Hz ±10 Hz = logic "0". 1270 Hz ±16 Hz = logic "1".

3.6.1.5.3 Power requirements. The transmitter shall operate on 20 ±2 Vdc and the maximum current drain shall not exceed 15 milliamperes (mA).

3.6.1.5.4 Clock frequency. The clock frequency of the transmitter shall be 12.0 ±0.1 Hz (±0.3 Hz during high temperature, low temperature, or humidity test conditions).

3.6.1.5.5 FSK tone frequencies. The FSK tone frequencies shall be 1270 ±16 Hz and 1070 ±10 Hz under all operating conditions.

3.6.1.5.6 Output level. The output level of the transmitter, as shipped, shall be -3.0 dBm minimum, 0.0 dBm maximum under all operating conditions.

3.6.1.5.7 Bit errors. The transmitter, when used with a data receiver, shall not generate any bit errors under any of the operating test conditions specified herein.

3.6.2 Receiver. The input of the receiver shall be by means of a 600 ohm balanced transmission line. The receiver shall decode each enciphered data word and set the appropriate alarm/status condition at the output. On receiving a unique word, the receiver shall initiate a synchronization cycle.

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3.6.2.1 Standards. Each receiver shall comply with two types of functional standards. One standard is based on receiver responses which are automatically set and which are dependent on the status of the data word itself. These status responses are listed in Table IV. The other standard is that the receiver shall respond with a status or alarm condition which corresponds to the status input to the transmitter as defined in Table V.

TABLE IV. Receiver responses to system faults

Event	Resulting receiver action
Five or more cipher bits are incorrect within a 25-second time period.	Alarm status shall be set.
A loss of word stream has persisted for 10 or more consecutive words (10 seconds or longer).	Alarm status shall be set.
Three or more parity errors are received in sequence.	A line fault error shall be set.
Five or more parity errors are detected within a 25-second time period.	A line fault error shall be set.
The level of the input signal to the receiver drops below the receiver demodulation threshold (dropout of baseband data occurs) for 1 second or longer.	A line fault error shall be set.

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TABLE V. Transmitter-input-status/receiver-output-status functional requirements

Condition at input of transmitter (condition must persist for three or more data words)	Resulting impedance with respect to designated pin and pin 5 of receiver, at output connector	Status indicated by this condition
1	2	3
Impedance between blue lead and gray lead at transmitter is more than 100k ohms.	Pin 1, 2 more than 100k ohms.	ALARM
Impedance between blue lead and gray lead at transmitter is less than 2k ohms.	Pin 1, 2 less than 2k ohms.	NO ALARM
Impedance between orange lead and gray lead at transmitter is more than 100k ohms.	Pin 1, 2 more than 100k ohms.	ALARM
Impedance between orange lead and gray lead at transmitter is less than 2k ohms.	Pin 1, 2 less than 2k ohms.	NO ALARM
Impedance between yellow lead and gray lead at transmitter is more than 100k ohms.	Pin 3 more than 100k ohms.	SECURE
Impedance between yellow lead and gray lead at transmitter is less than 2k ohms.	Pin 3 less than 2k ohms.	ACCESS
Impedance between white lead and gray lead at transmitter is more than 100k ohms.	Pin 4 more than 100k ohms.	AC POWER FAIL
Impedance between white lead and gray lead at transmitter is less than 2k ohms.	Pin 4 less than 2k ohms.	AC POWER ON

3.6.2.2 Code plug provisions. Provision shall be made to incorporate, in the receiver, a code plug which, when selected to match that in the transmitter, will enable the receiver to properly decode the data stream produced by the transmitter. This code plug, when installed, shall not be accessible without removing the receiver cover.

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3.6.2.3 Input connections. The input to the receiver shall be by means of pins 8 and 16 of the output connector, which shall connect to the 600 ohm transmission line from the transmitter.

3.6.2.4 Output connections. Table VI lists and identifies the output connections from the receiver.

TABLE VI. Output connections, receiver

Pin number of output connector	Function
1	ALARM/NO ALARM (Pins 1 and 2 interconnected).
2	ALARM/NO ALARM (Pins 1 and 2 interconnected).
3	ACCESS/SECURE OUTPUT.
4	AC POWER ON/AC POWER FAIL output.
5	Common return for status outputs.
6	Spare.
7	Spare.
8	Transmission line input.
9	20 Vdc input.
10	DC return.
11	Spare.
12	Spare.
13	Spare.
14	Spare.
15	Spare.
16	Transmission line input.

3.6.2.5 Function.

3.6.2.5.1 Power requirements. The receiver current drain shall not exceed 25 mA at 20 ±1 Vdc.

3.6.2.5.2 Clock frequency. The receiver clock frequency shall be 12.0 ±0.1 Hz (±0.3 Hz during high temperature, low temperature, or humidity test conditions).

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3.6.2.5.3 Bandwidth.

3.6.2.5.3.1 1270 Hz bandwidth. The 3 dB bandwidth of the receiver at constant output around the frequency of 1270 Hz, shall be 70 ±10 Hz.

3.6.2.5.3.2 1070 Hz bandwidth. The 3 dB bandwidth of the receiver at constant output, around the frequency of 1070 Hz, shall be 70 ±10 Hz.

3.6.2.5.4 Receiver sensitivity. The base band data shall not drop out until the transmission line attenuation is greater than -33 ±2 dBm.

3.6.2.5.5 Line fault sensitivity. The receiver shall have the line fault indicator illuminate when there is 2 ±1 dBm less attenuation in the transmission line than determined for base band data dropout in the sensitivity test.

3.7 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.8 Reliability. The data transmission group shall demonstrate a Mean-Time-Between-Failure (MTBF) of 10,000 hours or greater when tested as specified in 4.6.2.7.

3.9 Electromagnetic susceptibility. The electromagnetic susceptibility characteristics of the data transmission group shall conform to MIL-STD-461, Methods RS03, CS01, CS02, and CS06, as modified in 3.9.1 and 3.9.2.

3.9.1 Radiated. Radiated susceptibility shall be conducted in an electric field of 3 volts per meter (V/m) for both steady-state and keyed conditions of the EMI field generating equipment. This field applies to frequencies up to 500 megahertz (MHz). Above 500 MHz, the electric field shall be 1 V/m.

3.9.2 Conducted. Conducted interference signals shall be applied to both power and signal lines as follows:

(a) Continuous wave, 30 Hz to 400 kilohertz (kHz).

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(b) interfering spike voltage, 10 spikes 5 to 10 microseconds wide at a repetition rate of 7 to 10 pulses per second on input power leads only. Random phasing of the spikes with respect to the powerline frequency shall be employed.

3.10 Identification marking. The data transmission group shall be identified in accordance with MIL-STD-130, and the drawings referenced herein.

3.11 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9, except soldering.

3.11.1 Soldering. All soldering shall be performed in accordance with Requirement 5 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications, standards and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any test or examination by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the supplementary quality assurance provisions (SQAP) shown on the drawings. The drawings specify

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the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4.3 and 6.5).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction data transmission group shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The preproduction transmitters and receivers shall be subjected to the tests marked "X" in Column 1 of Table VII. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production data transmission groups will be selected at random by the Government from the data transmission groups being produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in Column 2 of Table VII to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production data transmission group shall not exclude the remaining data transmission groups from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production data transmission group to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4. shall be cause for rejection of the initial production data transmission group and shall

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be cause for refusal by the Government to continue acceptance of production data transmission groups until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on data transmission groups previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all data transmission groups accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, Inspection Level II. Sampling for tests shall be in accordance with MIL-STD-105, Inspection Level S-2.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined for the defects specified in 4.6.1. AQL shall be 2.5 percent defective.

4.5.3 Tests.

4.5.3.1 Individual. Each data transmission group shall be subjected to the test marked "X" in Column 3 of Table VII. Failure of the test shall be cause for rejection.

4.5.3.2 Samples. Samples selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in Column 4 of Table VII. AQL shall be 2.5 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The data transmission group shall be examined as specified herein for the following defects:

Major

101. Any part (or component) not in accordance with the SQAP requirement as shown on the drawings.
102. Components missing or not as specified.
103. Assembly not correct.
104. Interchangeability not as specified.

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105. Electromagnetic interference suppression components and network not identical to the approved model.
106. Identification or special markings missing or illegible.
107. Treatment and painting not as specified.
108. Workmanship not as specified.
109. Soldering not as specified.

4.6.2 Tests.

4.6.2.1 Test condition. Unless otherwise specified herein, tests shall be conducted at prevailing ambient temperature and humidity conditions. Prior to testing, each data transmission group shall be subjected to a 72-hour burn-in during which it shall be exercised through all its functions at a rate of nine times during each 24 hour period. All failures shall be recorded and analyzed for cause of failure. Corrective action shall be taken when appropriate. Copies of these failure reports shall be forwarded to the contracting officer. Units for testing shall be selected from those units having successfully completed the 72-hour burn-in.

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4.6.2.1 Thickness. Visible. The thickness of the visible [illegible]

4.6.2.2 Test schedule.

TABLE VII. Test schedule.

Preproduction	Initial production	Quality Conformance		Tests	Requirement paragraph	Test paragraph
		Individual	Sample			
1	2	3	4	5	6	7
X	X	X	X	Functional.	3.6	4.6.2.4
X	X	-	X	Low temperature.	3.5.1, 3.5.2	4.6.2.5.2
X	X	-	X	High temperature.	3.5.1, 3.5.2	4.6.2.5.3
X	X	-	X	Humidity.	3.5.1, 3.5.2	4.6.2.5.4
X	X	-	-	Shock	3.5.2	4.6.2.5.5
X	X	-	-	Vibration.	3.5.2	4.6.2.5.6
X	X	-	-	Electromagnetic susceptibility.	3.9	4.6.2.6
-	X	-	-	Reliability.	3.8	4.6.2.7

4.6.2.3 Test methods. Tests shall be conducted in accordance with the test method in the appendix and as specified herein. Calibration of test equipment shall be in accordance with MIL-C-45662.

4.6.2.4 Functional. The data transmission group shall be tested in accordance with the Appendix test method. Nonconformance to 3.6 shall constitute failure of this test.

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4.6.2.5 Environmental. The data transmission group shall be subjected to the following series of tests specified in MIL-STD-810.

4.6.2.5.1 Function. Subject the data transmission group to the tests specified in 4.6.2.4.

4.6.2.5.2 Low temperature. Low temperature Test Method 502.1, Procedure I, storage temperature -30° F for not less than 48 hours, equipment operating temperature -20° F.

4.6.2.5.3 High temperature. High temperature Test Method 501.1, Procedure II. Except that the storage temperature shall be +165° F; the operating temperature shall be +150° F.

4.6.2.5.4 Humidity. Method 507.1, Procedure I with the following exceptions:

(a) Step 5 shall be for 5 cycles (not less than 120 hours). Measurements at the completion of each cycle.
(b) During steps 2 and 3, the relative humidity shall be maintained at 95 percent or greater.
(c) During Step 4, the relative humidity shall be maintained at 92 percent or greater.
(d) Measurement shall commence 1 hour and 15 minutes prior to the completion of each 24-hour cycle. The cover of the data transmission group shall not be opened during measurements. The data transmission group may be removed from the environmental chamber for measurement; however, unsuccessful completion of measurements or unsuccessful return of the data transmission group to the test chamber within 1 hour and 15 minutes subsequent to the initiation of measurements shall constitute failure of the test.

4.6.2.5.5 Shock.

(a) Basic design: Conduct Method 516.2, Procedure I, equipment not operating, sawtooth shock pulse, peak value 20 g's, duration 11 milliseconds.
(b) Bench handling: Conduct Method 516.2, Procedure V.

4.6.2.5.6 Vibration. Method 514.2, Procedure X, Curve AX, equipment nonoperating.

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4.6.2.5.7 Failure criteria. Nonconformance to 3.5 shall constitute failure of these tests.

4.6.2.6 Electromagnetic susceptibility. The data transmission group shall be subjected to the tests of MIL-STD-461, as follows.

4.6.2.6.1 Radiation susceptibility. The data transmission group shall be tested in accordance with Test Method RS03. Electric field requirements of RS03 shall be modified and limited as specified in 3.9.1. All leads (sensor, alarm, and power) external to the enclosure shall be exposed to the electric field. The leads shall be not less than 4 feet in length.

4.6.2.6.2 Conducted susceptibility. The data transmission group shall be tested in accordance with Test Methods CS01, CS02, and CS06. The interference spike test specified in CS06 shall be modified as specified in 3.9.2.

4.6.2.6.3 Failure criteria. The contractor shall furnish the contracting officer a test plan and a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of these tests.

4.6.2.7 Reliability. Using the upper MTBF (θ_0) of 10,000 hours and a lower test MTBF (θ_1) of 5000 hours, the data transmission group shall be tested as specified herein with "accept" and "reject" criteria in accordance with MIL-STD-781, Test Plan VIIIC. Continue testing until either an "accept" or "reject decision is reached. A failure is defined as any malfunction which causes or may cause one or more of the following:

(a) Failure to commence operation, cessation of operation, or degradation of performance below the designated levels.
(b) Serious damage to the system.
(c) Serious personnel safety hazards.

4.6.2.7.1 Test procedure. One or more data transmitters and matching data receivers shall be randomly selected by a Government representative. The testing shall be conducted under ambient laboratory conditions, without temperature, humidity or vibration cycling. Operation shall be continuous throughout the test with all circuits exercised not less than nine times during each 24-hour period. The parameters to be measured shall be in accordance with Appendix test methods. Reliability tests shall commence after installation and a burn-in period of 72 hours.

4.6.2.7.2 Failure criteria. Nonconformance to 3.8 shall constitute failure of this test.

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4.7 Inspection comparison. The Government may select matching data transmitters and receivers at any time during the contract production period and subject these data transmission groups to the examination and the tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on data transmission groups selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial productior data transmission groups. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison data transmission group to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison data transmission group and shall be cause for refusal by the Government to continue acceptance of production data transmission groups until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on data transmission groups previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all data transmission groups accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

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110. Materials, methods, and containers not as specified for Level A or Level B. Each incorrect material, method, or container shall be considered one defect.
111. Individual preservation of components not as specified for Level A or Level B.
112. Code plugs not contained in matched sets as specified for Level A, Level B, or Commercial.
113. Boxed contents not cushioned as specified for Level A or Level B.
114. Packing not as specified for Level A or Level B.
115. Strapping not as specified for Level A.
116. Quantities packed together exceed the size or weight limitations of the box for Level A or Level B.
117. Marking missing, illegible, incorrect, or incomplete for Level A, Level B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Commercial as specified (see 6.2).

5.1.1 Level A. Each data transmitter and each data receiver shall be individually preserved in accordance with MIL-P-116, Method IIc. Each code plug shall be individually wrapped and cushioned and the matched set preserved together in accordance with MIL-P-116, Method IIc. The data transmitter and the data receiver shall each be placed in a close-fitting box conforming to PPP-B-636, W6c, style optional. The matched set of code plugs shall be placed in a close-fitting box conforming to PPP-B-566, Variety 1, style optional. Each item shall be cushioned within its box to prevent movement or damage.

5.1.2 Level B. Preservation shall be as specified in 5.1.1 for Level A, except Method IA-8 shall be used in place of Method IIc.

5.1.3 Commercial. Each data transmitter, data receiver, and matched set of code plugs shall be preserved in accordance with MIL-STD-1188. The matched set of code plugs shall be contained together to prevent loss or mismatch by the use of a plastic bag or paperboard box sealed to prevent accidental opening.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

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5.2.1 Level A.

5.2.1.1 Complete data transmission groups. Data transmitters, data receivers, and matched sets of code plugs of like quantity, preserved as specified in 5.1, shall be packed together to comprise complete data transmission groups, in a close-fitting box conforming to PPP-B-601, Overseas Type, Grade B, style optional, in quantities not to exceed the weight limitation of the box. Closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2) strapping shall be Finish A.

5.2.1.2 Individual components. When data transmitters or data receivers or matched sets of code plugs are procured separately they shall be packed as specified in 5.2.1.1, except only components of like description shall be packed within the same box.

5.2.2 Level B. Packing shall be as specified in 5.2.1.1 or 5.2.1.2 as applicable, except boxes shall be Domestic Type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Data transmitters, data receivers, and matched sets of code plugs preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129 and the special marking specified in 5.3.3.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188 and the special marking specified in 5.3.3.

5.3.3 Special marking. Each unit package containing a set of matched code plugs shall be marked in bold letters to indicate, "MATCHED SET. DO NOT ISSUE SEPARATELY". In addition, each shipping container shall be marked with code labels in accordance with MIL-STD-129. The labels shall contain the Project Code "JID".

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6. NOTES

6.1 Intended use. The data transmission group is intended for use as a component of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model, and number of models required (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(d) When initial production inspection is required and number of data transmission groups to be furnished (see 3.3).
(e) When code plugs are to be furnished a part of the data transmission group (see 3.6.1.3).
(f) Quantity of code plugs in matched set (see 3.6.1.3).
(g) Degree of preservation and packing required (see 5.1 and 5.2).
(h) When other than Finish B strapping is required (see 5.2.1.1 and 5.2.1.2).

6.3 Preproduction model. Any changes or deviations of production data transmission groups from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish data transmission groups conforming to this specification.

6.4 Definitions. The following definitions apply throughout this specification.

6.4.1 False alarm. An alarm generated by stimuli other than those for which the data transmission group is designed to alarm.

6.4.2 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, or change of fit of any component; physical change

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which impairs the mechanical integrity of the data transmission group; evidence of delamination or water penetration into integrated circuits, printed wiring boards, external components, or any physical condition resulting in noncompliance of the units with the provisions of this specification.

6.4.3 Supplementary quality assurance provisions (SQAP). A SQAP is a contractual requirement that supplements Section 4 of the specification. SQAP's indicate the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of the repair parts.

6.5 Supplementary quality assurance provisions (SQAP) records. The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of SQAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each data transmission group.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of data transmission groups.

6.8 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

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Army - ME
Air Force - 99

Preparing activity:
Army - ME

Project 6350-0057

Review activity:
DLA - GS

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APPENDIX

FOR

TRANSMITTER, DATA

AND

RECEIVER, DATA

10. SCOPE

10.1 Scope. The test methods contained in this appendix are to determine whether data transmission groups procured under this specification conform to the requirements set forth.

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TEST METHODS

1. TESTS TO BE PERFORMED:

(a) Transmitter power requirements.
(b) Transmitter clock frequency.
(c) Transmitter FSK tone frequency and amplitude.
(d) Output level.
(e) Receiver power requirements.
(f) Receiver clock frequency.
(g) Receiver bandwidth test.
(h) Synchronization check.
(i) Sensitivity test.
(j) Low level data receiver input line fault check.
(k) Transmitter/receiver status transmission verification.
(l) Receiver transmission line fault operation verification.
(m) Receiver parity test.
(n) Cipher error test.
(o) Bit error test.
(p) Open line synchronization test.

2. TEST APPARATUS:

(a) Power supply A - Power Design X-MOD-10, or equivalent.
(b) Voltmeter - Hewlett-Packard 34740A Display/34702A multimeter, or equivalent.
(c) Ammeter - Simpson 260 multimeter, or equivalent.
(d) Frequency counter - Hewlett-Packard 5532A, or equivalent.
(e) Test fixture - see Figures A-1 and A-2.
(f) Variable attenuator pad (600 ohm) - Hewlett-Packard 350D, or equivalent.
(g) Stopwatch.
(h) Power supply B, 20 $\pm$2 Vdc, battery operated.
(i) Signal generator - audio frequency, 600 ohm output.

3. TEST PREPARATION:

(a) Set power supply output voltages to 20 Vdc.
(b) Turn off power supplies.
(c) Remove covers from both data transmitter and data receiver. Install one of a matched set of code plugs in each unit.

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(d) Connect data transmitter, data receiver, and associated test equipment as shown in Figure A-1.
(e) Turn on power supplies and measure output voltages to ensure that they are set for 20 volts. Ensure that switch S8 is in the CLOSED position.

4. TEST PROCEDURE:

(a) Transmitter power requirements.

(1) Disconnect data transmitter red wire from pin 18 of tester.
(2) Increase power supply B voltage to 22 Vdc.
(3) Connect ammeter from transmitter red wire to pin 18 of tester. Current to data transmitter must be 15 mA or less.
(4) Set power supply B to 18 Vdc. Current to data transmitter must be 15 mA or less.
(5) Remove meter and reconnect transmitter red wire to pin 18 of tester.
(6) Readjust power supply B voltage to 20 Vdc.

(b) Transmitter clock frequency.

(1) Measure clock frequency at TP2 (red) of transmitter printed wiring board (PWB) A2. Frequency shall be 12.0 $\pm$0.1 Hz ($\pm$0.3 Hz during high temperature, low temperature, or humidity test conditions).

(c) Transmitter FSK tone frequency and amplitude.

(1) Ground TP5 (green) of transmitter PWB A1.
(2) Measure frequency at TP7 (violet) of transmitter PWB A1. Frequency shall be 1270 Hz $\pm$16 Hz under all operating conditions.
(3) Connect TP5 (green) of transmitter PWB A1 to E14.
(4) Measure frequency at TP7 (violet) of transmitter PWB A1. Frequency shall be 1070 Hz $\pm$10 Hz under all operating conditions.
(5) At TP1, (brown) of transmitter PWB A1, measure the output level. It shall be between 0.0 and -3.0 dBm.

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(d) Output level.

(1) Connect a root mean square (rms) voltmeter across TB2-8 and TB2-9 of the test set.
(2) Verify that the output level of the transmitter is between 0.0 and -3.0 dBm.

(e) Receiver power requirements.

(1) Disconnect Data Receiver wire (red) connected to terminal 1 of tester.
(2) Readjust power supply A voltage to read 21 volts.
(3) Connect ammeter from loose wire to terminal 1 of tester. Input current to data receiver must be 25 mA or less.
(4) Readjust power supply to 19 Vdc. Input current must be 25 mA or less.
(5) Remove meter and reconnect loose wire to terminal 1 of tester.
(6) Readjust power supply A voltage to 20 volts.

(f) Receiver clock frequency.

(1) Place switch S8 to open position (opens transmission line).
(2) Measure frequency at TP6 (blue) of PWB A2 of receiver. Frequency shall be 12.0 $\pm$0.1 Hz ($\pm$0.3 Hz during high temperature, low temperature, or humidity test conditions).

(g) Receiver bandwidth.

(1) Connect an audio generator to J7 and J8 of test set.
(2) Connect an alternating current (ac) voltmeter to TP5 (green) of PWB A4 of the Data Receiver.
(3) Close switch S8 (transmission line).
(4) Set audio generator to 1170 Hz $\pm$10 Hz and adjust output to -30 dBm.
(5) Adjust transformer T1 for a peak reading of the ac voltmeter.
(6) Connect an ac voltmeter to TP2 (red) of PWB A4.
(7) Tune audio generator until ac voltmeter peaks (approximately 1270 Hz) and adjust output of generator until input at J7-J8 reads -32 dBm.

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(8) Verify that the peak frequency is 1270 ±20 Hz. If not, adjust R21. Verify that the signal at TP2 is not saturated by decreasing the input at J7-J8 by 3 dB noting that the ac voltmeter reading decreases by 3 dB. If not, decrease input and repeat above.
(9) Adjust the frequency of the audio generator above and below the peak frequency, keeping the output constant, until the ac voltmeter reading decreases by 3 dB.
(10) Verify that the 3 dB bandwidth is 70 ±10 Hz.
(11) Connect an ac voltmeter to TPI (brown) of PWB A4 of the data receiver.
(12) Repeat (7), (8), (9), and (10) except peak at 1070 Hz. Adjust R15 to peak at 1070 Hz.

(h) Synchronization check.

(1) With system operating, turn power supply B off for 5 seconds.
(2) Turn power back on and verify automatic system synchronization within 15 seconds (LED 1, LED 2, and LED 3 continuously on).
(3) Turn off power supply A for 5 seconds.
(4) Turn power supply A back on and depress PB1 to initiate a RESYNC cycle. System will synchronize and remain synchronized for at least 2 minutes (LED 1, LED 2, and LED 3 continuously on).
(5) Set switch S3 to ON (LED shall turn off).
(6) Depress PB1 to initiate RESYNC cycle. (LED 2 shall momentarily turn on and then return to off state thereby verifying receipt of RESYNC word).
(7) Return switch S3 to off position.

(i) Sensitivity test.

(1) Insert the variable attenuator pad (600 ohms) into transmission line.
(2) Resync system (press PB1 and wait 15 seconds).
(3) Set the variable attenuator such that the input level to the receiver at TB2-8 and TB2-9 is -30 dBm (line fault indication may result).
(4) Reset line fault LED. (If line fault LED will not reset, see test (j).)
(5) Allow system to operate for 2 minutes.

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(6) Connect oscilloscope to TP8. Observe the base band data. Increase attenuator until the base band data at TP8 drops out (line fault shall result).
(7) Record level. Signal level at which dropout of base band data occurs shall be -33 ±2 dB.

(j) Low level data receiver input line fault check.

(1) Set attenuator pad such that input level to receiver is -30 dB.
(2) Depress PB1 to initiate a RESYNC cycle.
(3) Depress line fault reset switch.
(4) Increase attenuation until line fault indicator illuminates. Line fault indicator shall have illuminated at 2 ±1 dB less attenuation than for base band data dropout as determined in sensitivity test. If the line fault indicator does not illuminate, adjust R35 on PWB A4 to obtain correct value.
(5) Remove variable attenuator pad from transmission line.

(k) Transmitter/receiver status transmission verification.

(1) Assure that switches S1, S2, S3, and S4 are in the OFF (down) position and that system is synchronized (LED 1, LED 2, LED 3 on).
(2) Turn switch S1 ON (instantaneous alarm initiation).
(3) LED 1 (alarm indication) should turn OFF.
(4) Return switch S1 to the OFF (no alarm) position and wait until LED 1 is again lighted.
(5) Turn switch S2 ON (latched alarm initiation).
(6) LED 1 should turn OFF.
(7) Return switch S2 to the OFF position and wait until LED 1 is lighted.
(8) Turn switch S3 ON (access initiation).
(9) LED 2 (access indication) should turn OFF.
(10) Return switch S3 to the OFF position and wait until LED 2 is lighted.
(11) Turn switch S4 ON (ac power fail initiation).
(12) LED 3 (ac power fail indicator) should turn OFF.
(13) Return switch S4 to the OFF position and wait until LED 3 is lighted.

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(14) Turn on switches S1, S2, S3, and S4 simultaneously. LED 1, LED 2, and LED 3 should all turn off simultaneously within approximately 3-5 seconds.
(15) Turn off switches S1, S2, S3, and S4 simultaneously. LED 1, LED 2, and LED 3 should all turn on simultaneously within approximately 3-5 seconds.

(l) Receiver transmission line fault operation verification.

(1) Open transmission line switch (S8).
(2) Verify that the line fault indicator on the Data Receiver lights within 0.8 and 1.2 seconds.
(3) Verify that LED 1 turns OFF within 5 seconds.
(4) Close switch S8, restoring transmission line continuity. Reset line fault on Data Receiver. Assure that the line fault indicator goes OFF.
(5) Short transmission line by connecting J7 and J8 to ground.
(6) Verify that the line fault indicator on the Data Receiver lights within 0.8 and 1.2 seconds.
(7) Verify that LED 1 turns OFF within 5 seconds.
(8) Remove short from between J7 and J8 and ground. Resynchronize by pressing PB1 and waiting 15 seconds before continuing.

(m) Receiver parity test.

(1) Connect J3 on tester to TP4 (yellow) on PWB A1 on the Data Transmitter.
(2) Ensure that switch S7 is in the OFF position and press the line fault reset on the Data Receiver.
(3) Connect J1 on tester to TP7 (violet) on PWB A3 of the Data Receiver.
(4) Ensure that switch S5 is in the ON position and press PB2. LED 4 should be lighted.
(5) Turn switch S7 ON and watch to see that LED 5 lights. When LED 5 lights, immediately turn switch S7 OFF. LED 4 should light.

(NOTE: Insure that only two parity errors are generated by data receiver.)

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(6) Verify that line fault does not occur.
(7) Wait for 30 seconds. Press PB2.
(8) Turn switch S7 ON. LED 5 should light, then LED 4 again. When LED 4 lights for the second time, switch S7 should be turned OFF. LED 5 should light for the second time. (Three consecutive parity errors).
(9) Verify line fault occurrence.
(10) Wait 30 seconds. Reset line fault. Press PB2.
(11) Turn switch S7 ON. Start stopwatch when LED 5 lights.
(12) When LED 5 lights, turn switch S7 OFF. LED 4 will light.
(13) At the 5 second mark on stopwatch, turn switch S7 ON. When LED 5 lights, turn switch S7 OFF. LED 4 should light.
(14) At 22 seconds on stopwatch, turn switch S7 ON.
(15) When LED 5 lights, turn switch S7 OFF. LED 4 will light.
(16) Verify line fault indicator lights.
(17) Reset line fault and stopwatch. Push PB 2. Wait 30 seconds.
(18) Turn switch S7 ON. Start stopwatch when LED 5 lights.
(19) When LED 5 lights, turn switch S7 OFF. LED 4 will light.
(20) At the 5 second mark on stopwatch, turn switch S7 ON. When LED 5 lights, turn switch S7 OFF. LED 4 should light.
(21) At 25 seconds on stopwatch, turn switch S7 ON.
(22) When LED 5 lights, turn switch S7 OFF. LED 4 will light.
(23) Verify line fault indicator does not light.
(24) Remove test probes from data transmitter and receiver.

(n) Cipher error test.

(1) Connect J3 on tester to TP2 (red) on PWB A1 of Data Transmitter.
(2) Connect J4 on tester to TP3 (orange) on PWB A1 of Data Transmitter.
(3) Connect J1 on tester to TP3 (orange) on PWB A2 of Data Receiver.
(4) Ensure that switches S6 and S7 are in the OFF position and press PB2.
(5) Turn switches S6 and S7 ON. Start stopwatch when LED 5 lights.
(6) Allow LED 5 to light, then LED 4 and LED 5. When LED 5 lights the second time, turn switches S6 and S7 OFF. LED 4 will light again.
(7) At 22 seconds on stopwatch, turn switches S6 and S7 ON.
(8) When LED 5 lights, turn switches S6 and S7 OFF. LED 4 will light.

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(9) LED 1 should go out then come back on.
(10) Press push-button 2. Wait 30 seconds.
(11) Turn switches S6 and S7 ON. Start stopwatch when LED 5 lights.
(12) Allow LED 5 to light, then LED 4 and LED 5. When LED 5 lights for the second time, turn switches S6 and S7 OFF. LED 4 should light.
(13) At the 25 second mark on stopwatch, turn switches S6 and S7 ON.
(14) When LED 5 lights, turn switches S6 and S7 OFF. LED 4 will light.
(15) LED 1 should not go OUT.
(16) Remove test connections from data receiver and transmitter.
(17) Turn off power to data receiver and transmitter.

(o) Bit error test.

(1) Connect J2 of the bit error test circuit to TP3 (orange) on PWB A3 of data receiver.
(2) Connect J1 of the bit error test circuit to TP7 (violet) on PWB A3 of data receiver.
(3) Apply power to data receiver and transmitter and re-synchronize system.
(4) Press PB2 on bit error test circuit and allow units to operate for 5 minutes.
(5) Note that neither LED 5 nor LED 7 on the bit error test circuit is lighted.

(p) Open line synchronization test.

(1) Verify that the system is still in synchronization (LED 1, LED 2, and LED 3 are continuously on).
(2) Open the transmission line (switch S8) for 10 seconds. LED 1 will go OUT within 5 seconds. LED's 2 and 3 should remain ON.
(3) Close the transmission line and the system should remain in synchronization. LED 1 should come ON.

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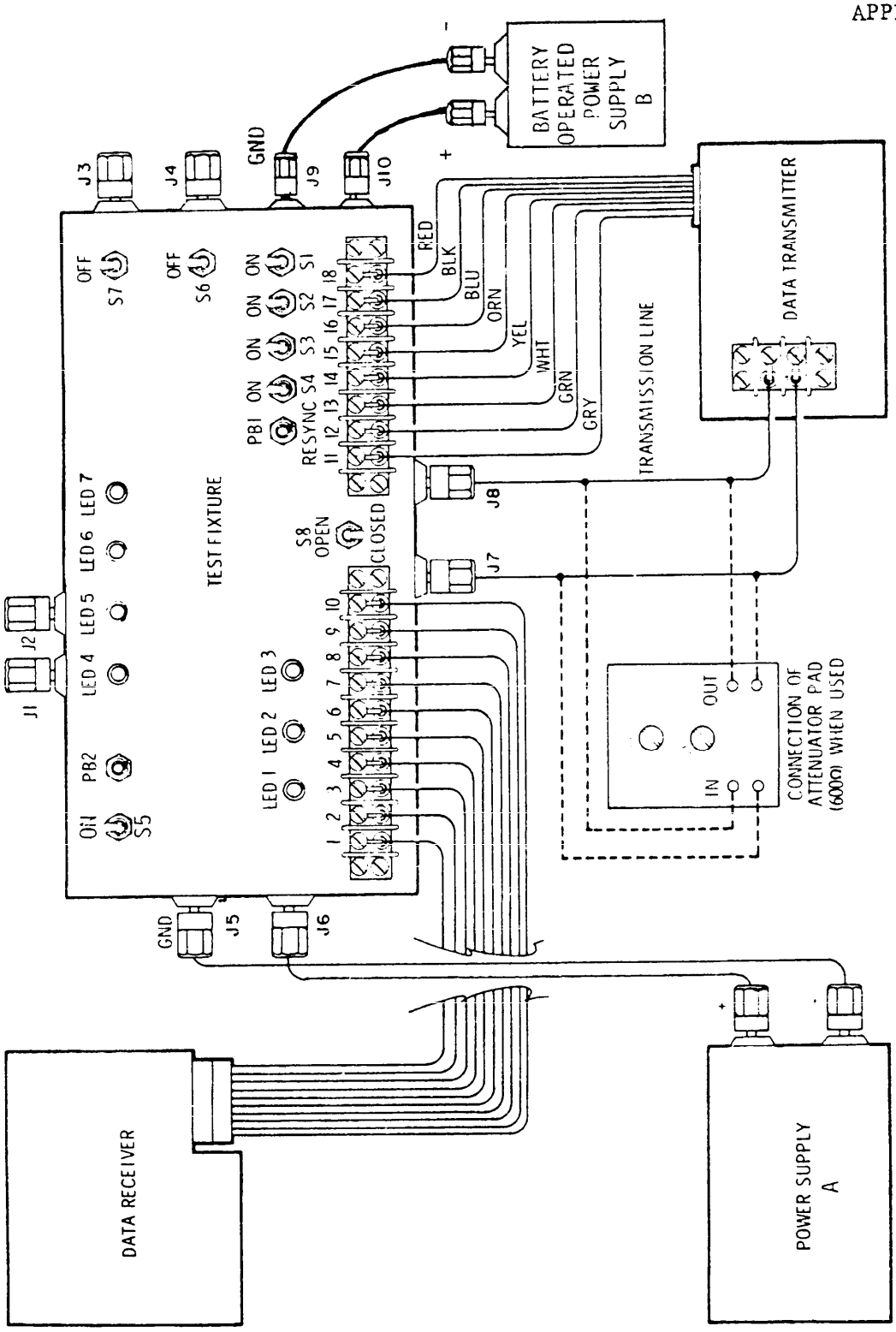


FIGURE A-1. Test fixture and test setup.

X-3157B

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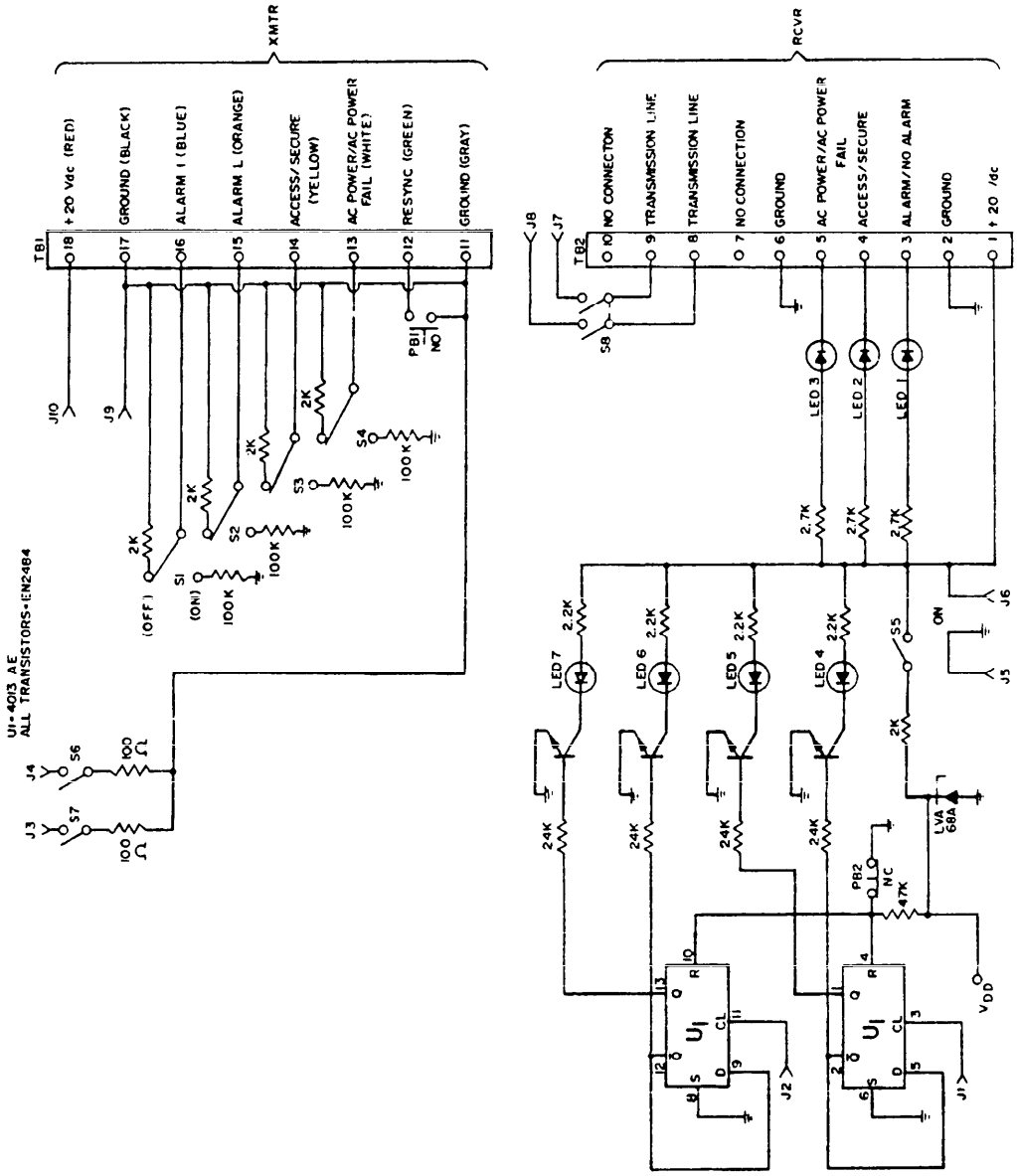


FIGURE A-2. Test fixture schematic diagram.

X-3158D

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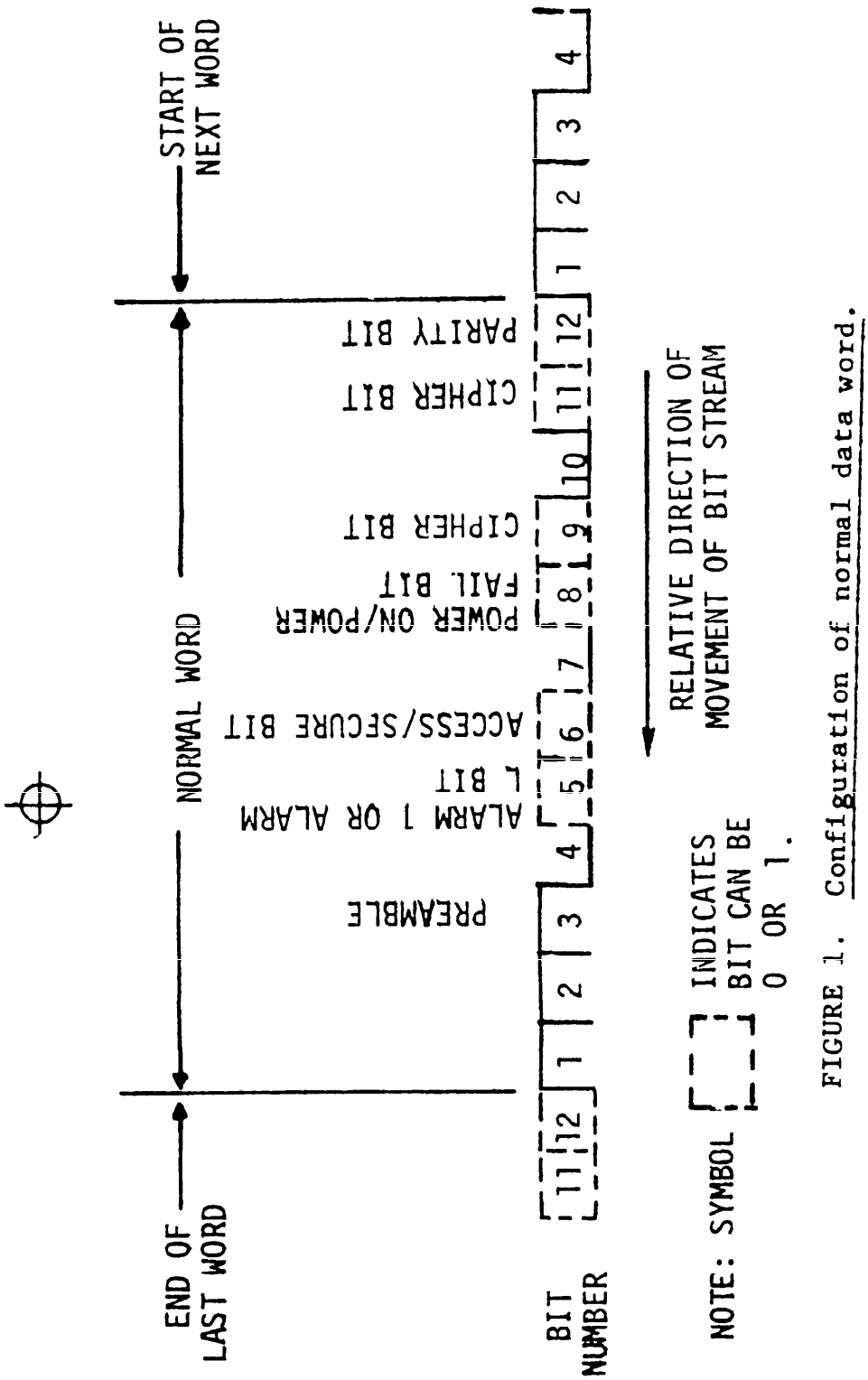


FIGURE 1. Configuration of normal data word.

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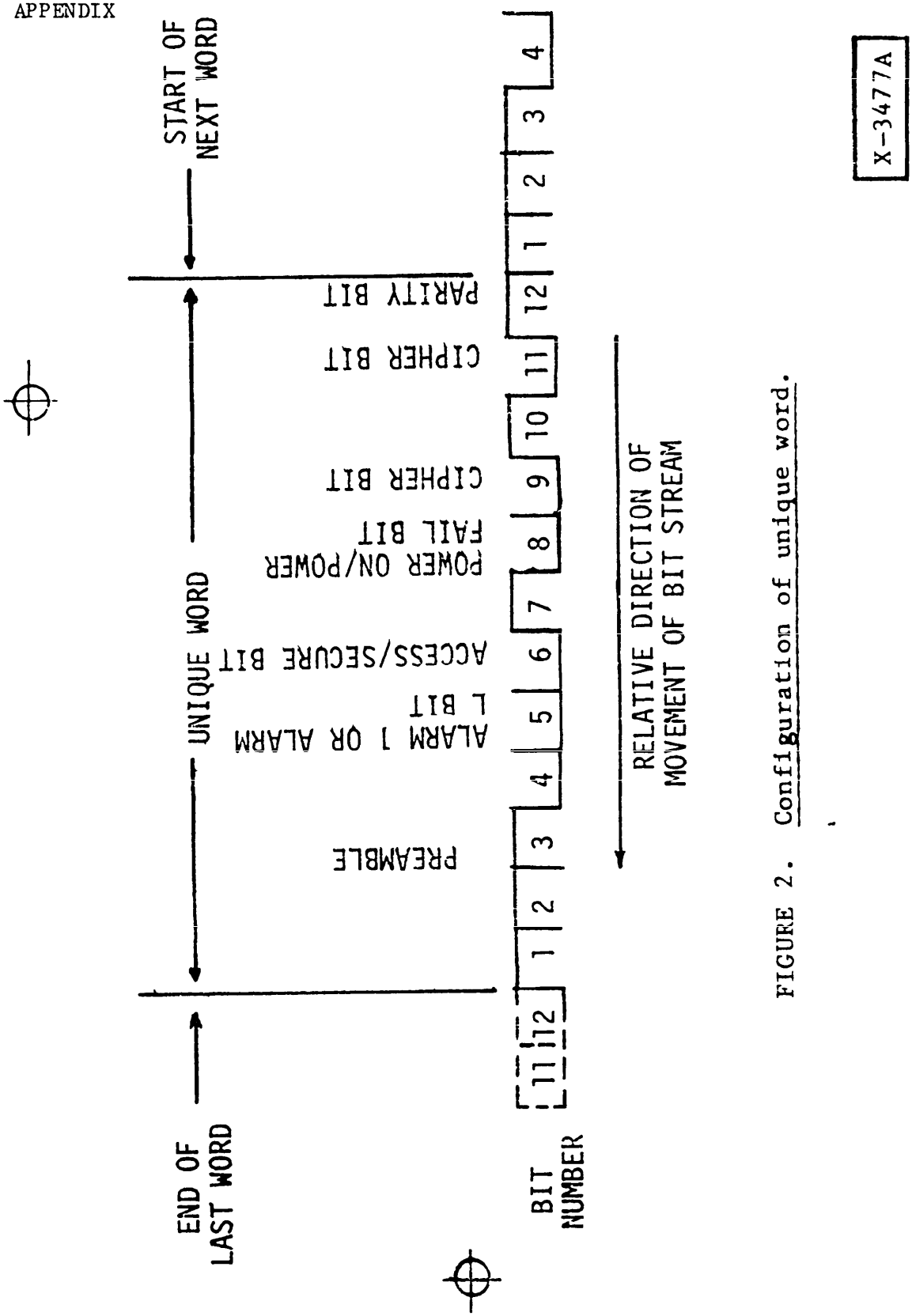


FIGURE 2. Configuration of unique word.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE: MIL-T-52877B Transmission Group Data: Transmitter, Data: T-1257/FSS-9(V) and Receiver, Data: R-1861/FSS-9(V)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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